

# LECTURE 9:

## Cooperation in MAS (V): implicit methods

**Introduction to Multi-Agent Systems (MESIIA, MIA)**

**Department of Computer Science and Mathematics**

**Rovira i Virgili University**

**Fall 2019**

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# Outline of the lecture

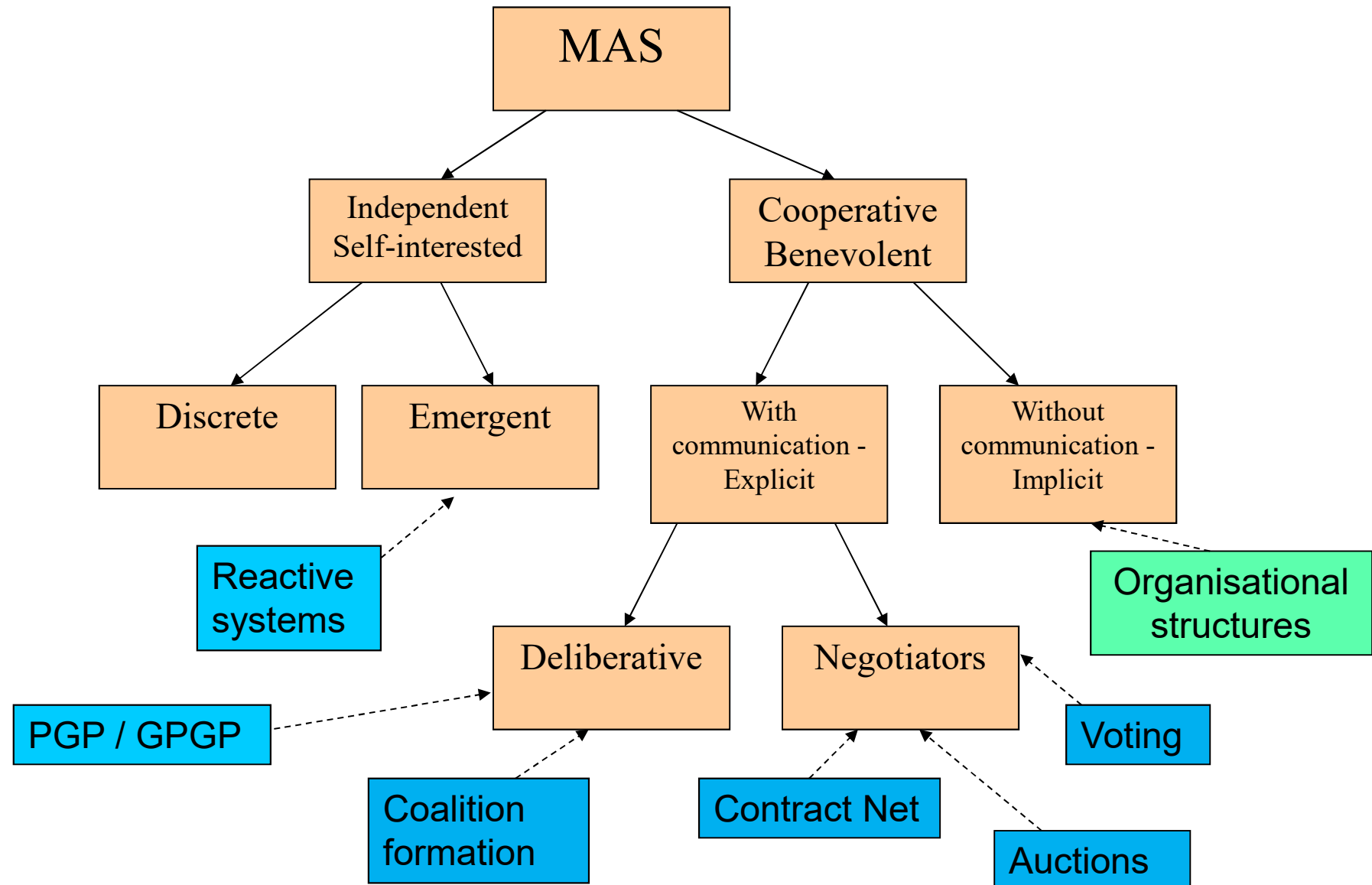
- Implicit cooperation in MAS
    - Indirect cooperation
    - Societal views of MAS
      - Electronic institutions
      - Organisational structures
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# Coordination [recall past lectures]

- An **activity** is a set of potential operations an **actor** (an agent playing a certain role) can perform, with the aim of achieving a given goal
  - **Coordination** could be defined as the process of **managing dependencies between activities**. By such process an agent **reasons** about its local actions and the foreseen actions that other agents may perform, with the aim to make the community to **behave globally in a coherent manner**
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# Cooperation hierarchy [last lectures]



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# Implicit cooperation

- A group of distributed cooperative agents behaves in a socially coordinated way in the resolution of a global problem **without an explicit exchange of communication messages**
  - In many cases the **environment** acts as the (indirect) interaction mechanism
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# Motivation (I)

- ❑ **Speed**: it takes too long to communicate with others – by then the opportunities are missed
  - E.g. Football game – simple signals may work, but lengthy explanations don't...
  - In general, very dynamic environments
- ❑ **Security**: not wanting others to know what your plans are



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# Motivation (II)

- ❑ **Complexity**: some agents may be too simple to deal with the task of generating and understanding long plans
    - Reactive rule-based robots
    - Reasoning involved in **Partial Global Planning** or **coalition formation** algorithms
  - ❑ **Lack of a communication channel**: there may actually be no way to communicate
    - Physical robots with limited communication range
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# Options for implicit cooperation

1. Indirect cooperation through the effects on the environment of the actions of each agent
  2. Reasoning mechanisms (at the individual or societal levels)
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# 1-Emergent Behaviour [recall past lectures]

- Coordination in cases where:
    - There is no communication between agents
    - There is no mechanism for enforcing a-priori social rules / laws
    - Agents have their own agenda/goals
  - The resulting coherent global behaviour is emergent and cannot be said to be based on joint plans or intentions
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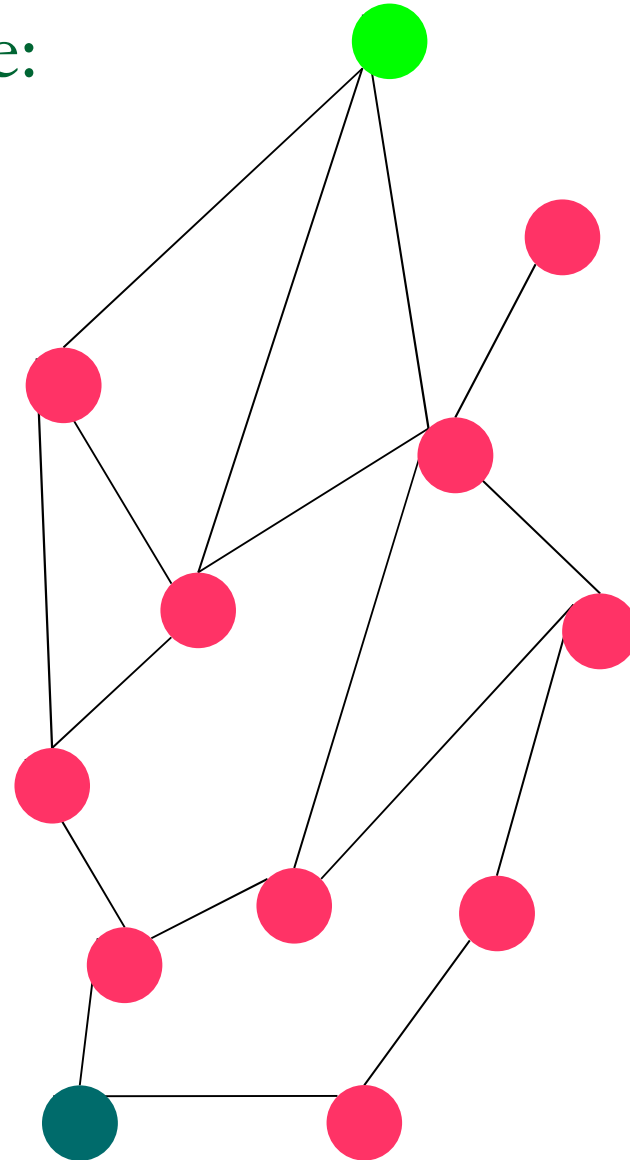
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## Basic difference

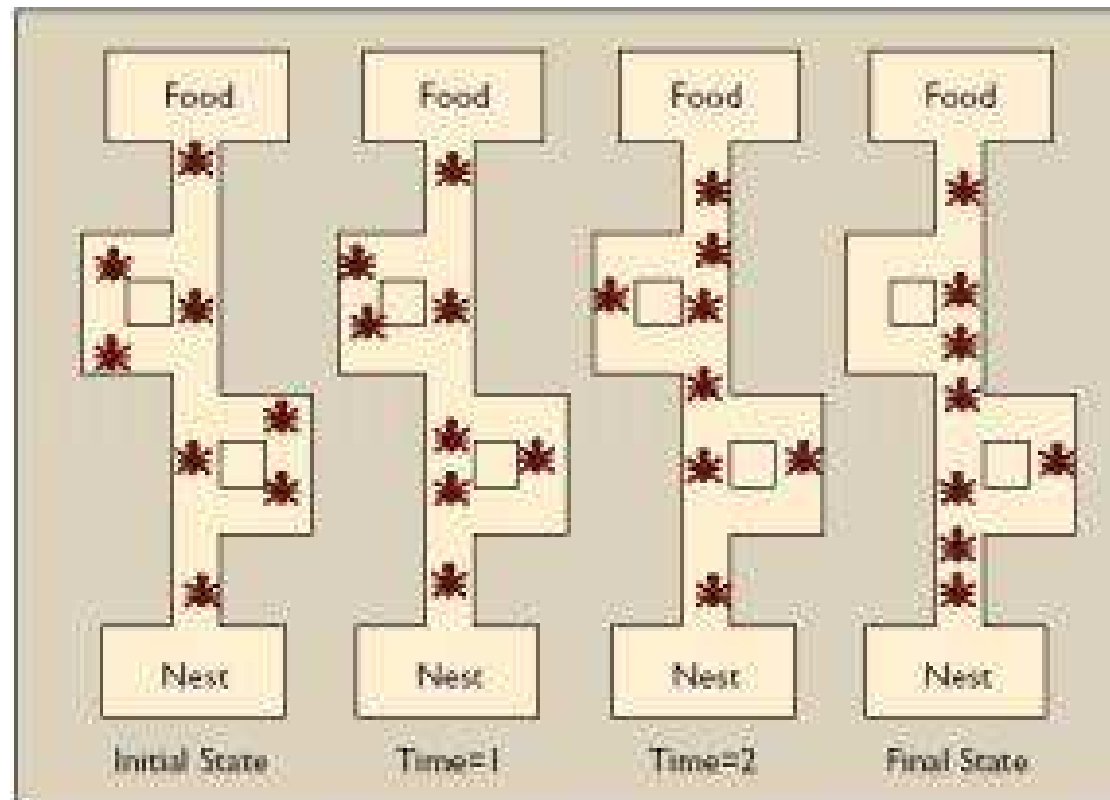
- **Emergent coordination**: agents are self-interested, they do not care about the other agents in the system, there isn't any high level design of the emergent behaviour
  - **Implicit coordination** (also giving rise to emergent coordinated global behaviour): although agents do not communicate with each other, the **designer** of the system intends to **provoke the emergence** of the socially intelligent problem solving activities
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# Implicit coordination example: Network Routing

- Network Routing problems are challenging. Solutions need to be:
  - Dynamic
  - Robust
- Network of  $N$  nodes,  $L$  links. Traffic flows as packets traverse the network
- There are protocols that compute cumulative shortest path measures

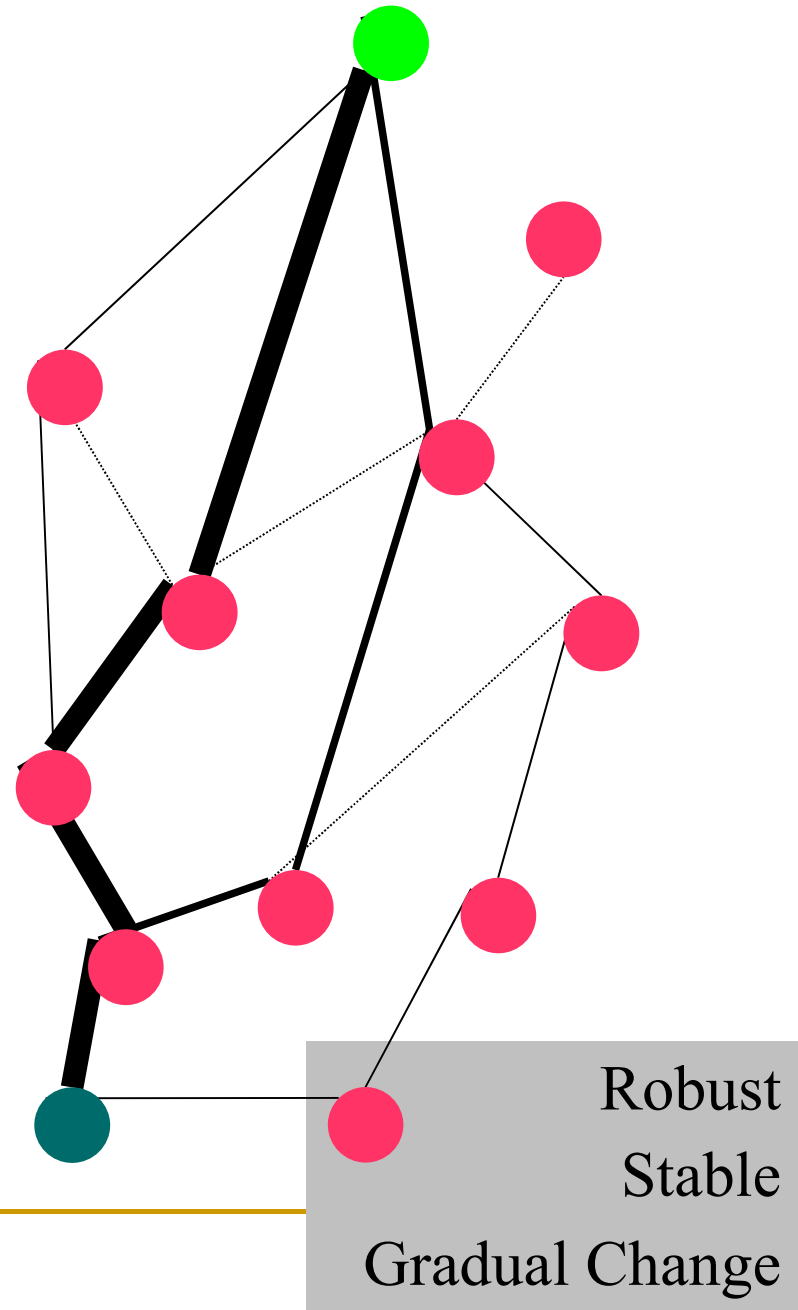


# Ants discover shortest paths



# Network Ants

- Ants randomly explore the network until they find a specific node
  - They mark the traversed paths with “pheromone”
- Ants seeking destinations tend to follow pheromone trails
- Pheromones degrade over time



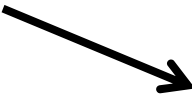
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# Pheromone tables

- Each node contains a **table of probabilities** (*pheromone table*) for each possible destination in the network
    - In a 30-nodes network, each node keeps 29 tables
  - The entries on the tables are the probabilities which influence the ants' selection of the next node on the way to their destination node
  - Pheromone laying = updating probabilities
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# Pheromone tables example

- A network with 6 nodes, node 1 is connected with nodes 2, 4 and 5.
- The pheromone tables in node 1 would look like this
- For instance, if an ant arrives at node 1 and wants to go to node 3, the most probable route is through node 4 (but it may also decide to go through nodes 2 or 5)

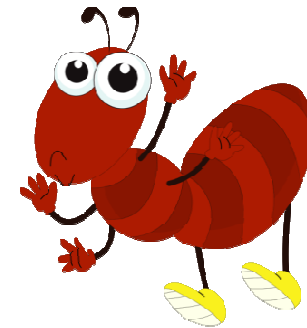


|                     |   | Next node |      |      |
|---------------------|---|-----------|------|------|
|                     |   | 2         | 4    | 5    |
| Destination<br>node | 2 | 0.90      | 0.05 | 0.05 |
|                     | 3 | 0.25      | 0.60 | 0.15 |
|                     | 4 | 0.10      | 0.85 | 0.05 |
|                     | 5 | 0.10      | 0.10 | 0.80 |
|                     | 6 | 0.40      | 0.30 | 0.30 |

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## Simulation (I)

- Pheromone tables are initialized with random values
- At each step, ants can be launched from any node in the network, with a **random destination node**
- Ants move from node to node, selecting the next node to move to according to the **probabilities in the pheromone tables** for their destination node



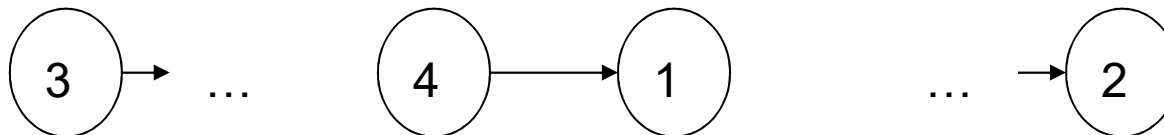
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## Simulation (II)

- When ants arrive at a node, they **update the probabilities** of that node's pheromone table entries corresponding to their **source** node
  - They alter the table to **increase the probability** pointing to their previous node
  - Ants moving away from their source node can only directly affect those ants for which it is the destination node
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# Pheromone laying example

- An ant has to go from node 3 to node 2; in the way, it travels from node 4 to node 1
  - First, it modifies the table in node 1 corresponding to destination node 3, increasing the probability of selecting the link to node 4
  - After that, it selects the next node randomly according to the probabilities of the table in node 1 corresponding to the destination node 2



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# Increasing/decreasing pheromones

- Pheromones are increased with the following formula

$$p = (p\_old + \Delta(p)) / (1 + \Delta(p))$$

- As all the entries must add up to 1, the other entries have to be decreased as follows

$$p = p\_old / (1 + \Delta(p))$$

- Note that probabilities may never be 0
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# Basic ideas of implicit cooperation

- Agents **do not talk to each other** directly
  - Agents can **modify the environment**, and these modifications influence the behaviour of the other agents in the system
    - These modifications have been carefully designed
  - All the agents contribute towards a **useful global behaviour** of the community
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## 2-Reasoning mechanisms for coordination

- A) Thinking about individual agents
    - Methods that allow building a model of the other agents of the system
  - B) Thinking about the whole agent's society
    - Methods that try to impose some kind of rules/laws/structure/organisation in the multi-agent system
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# A-Agent Modelling

## ■ Recursive Modelling Methods

- Assume the others have a similar structure to you – and may have a model of you...
- Try to deduce their beliefs/desires/intentions from their actions on the environment

## ■ Plan Recognition

- Analyse the sequences of activities of other agents and try to discover their **plans** (and, from them, identify the potential end goals of their actual actions)

## ■ Game Playing / Game Tree Search:

- Modelling opponents (e.g. minimax search)
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# B-Thinking about Society

- Common approaches include:
  - **Social Laws:** global rules which agents follow and lead to “coherent behaviour”, either instilled in the agent or communicated when entering the environment (e.g. - “driving on the right hand side”)
  - Electronic Institutions
  - Organisational structures

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# Institutions

- **Institutions** are a kind of social structure where a corpora of **constraints** shape the behaviour of the members of a group
  - The definition of a (human) **Institution** usually includes:
    - **Norms** about the interactions
    - **Procedures** and **protocols** to be followed
    - **Conventions**: acceptable (and unacceptable) actions within the institution
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# e-Institutions

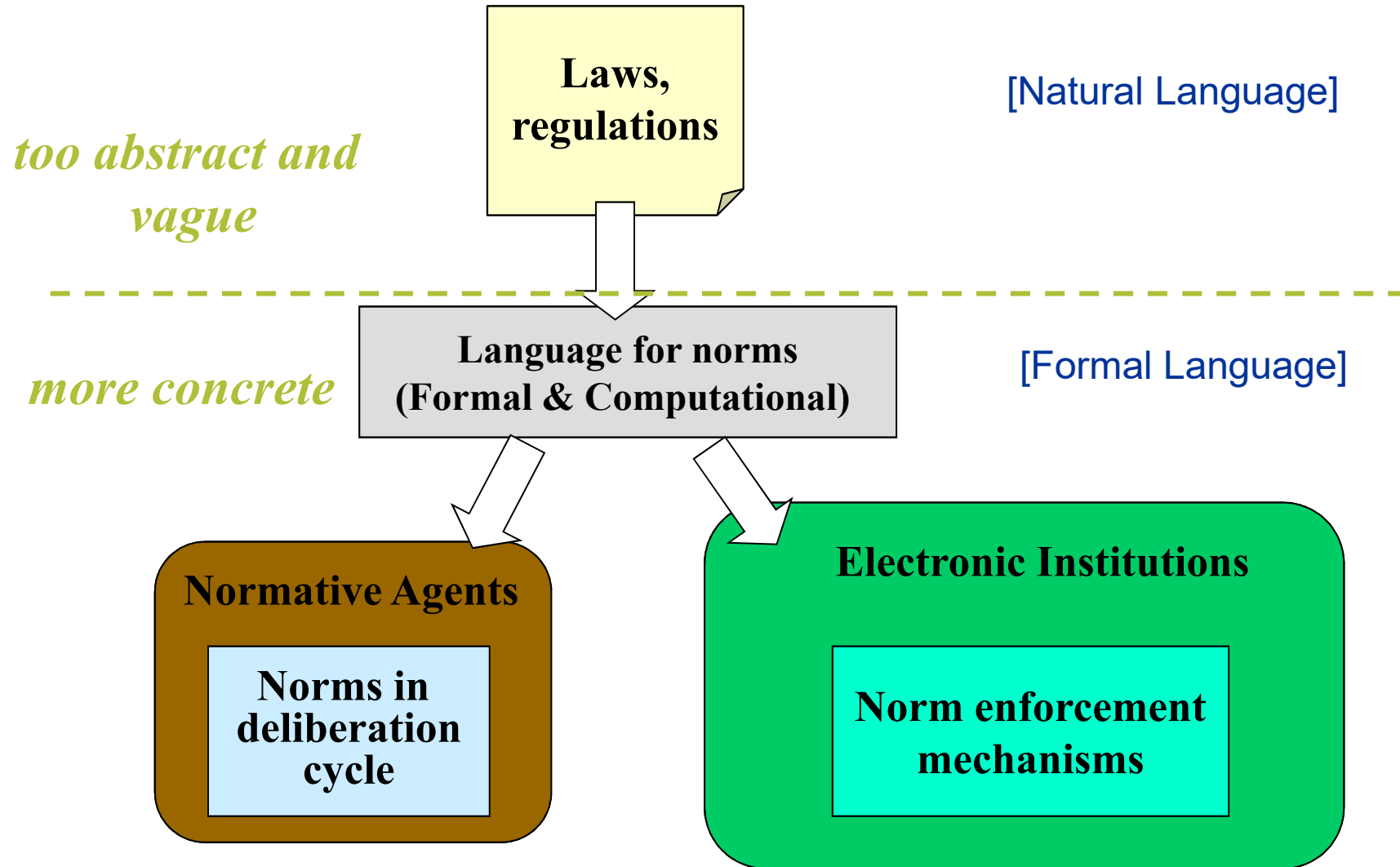
- An **e-Institution** is the computational model of an institution through
    - The specification of the institution's *norms* in some suitable formalism
    - The formal specification of the institution's admissible **procedures and protocols**, which follow the established **conventions**
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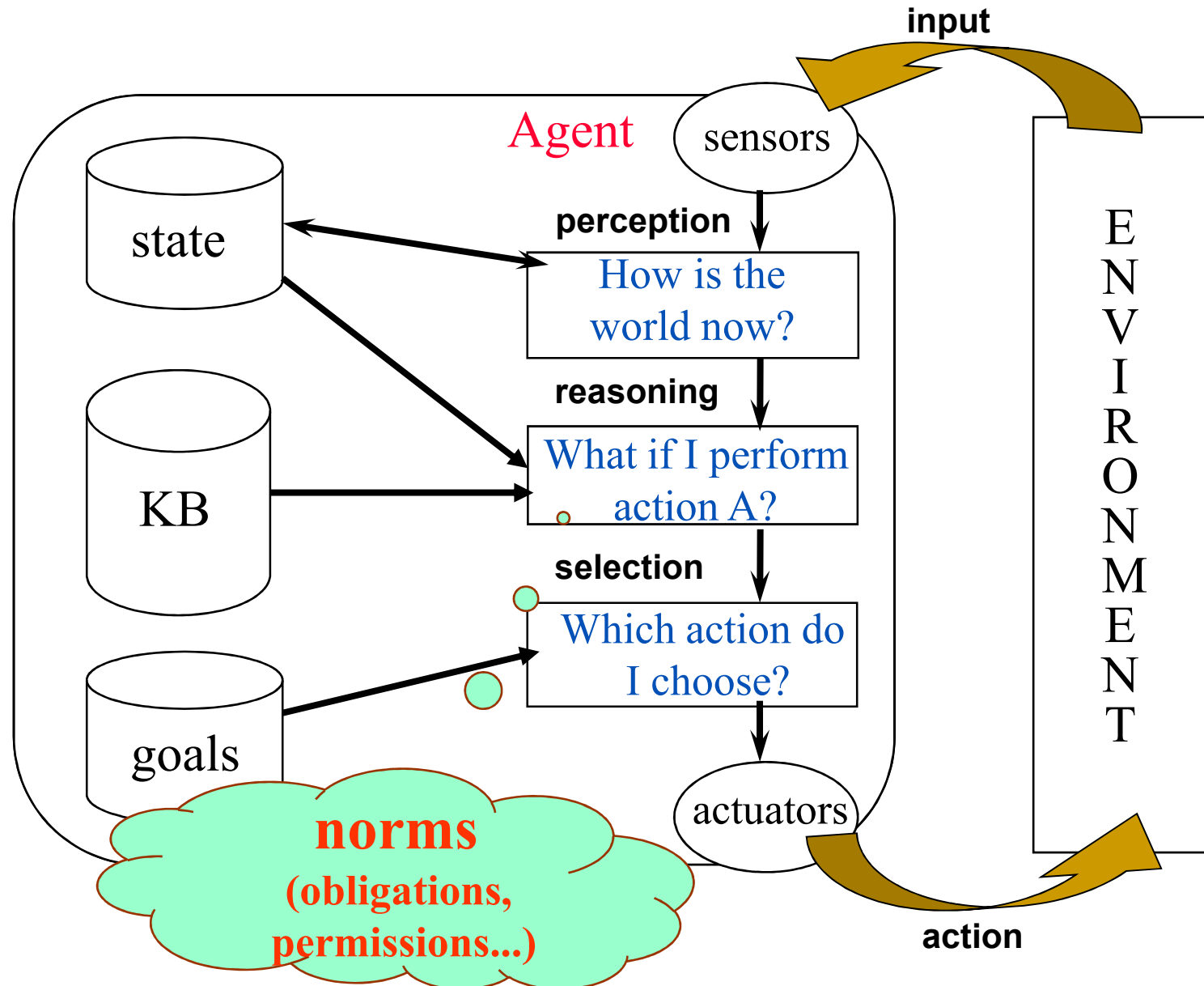
# e-Institutions and MAS

- In the context of MAS, e-institutions:
    - ❑ reduce **uncertainty** about other agents' behaviour
    - ❑ allow agents to **foresee the outcome** of an interaction
    - ❑ simplify the **decision making process** (by reducing the possible future actions)
  - ❑ **Agent behaviour guided by Norms**
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# Why a Language for Norms?



# Influence of norms in the BDI deliberation cycle



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# AMELI (I)

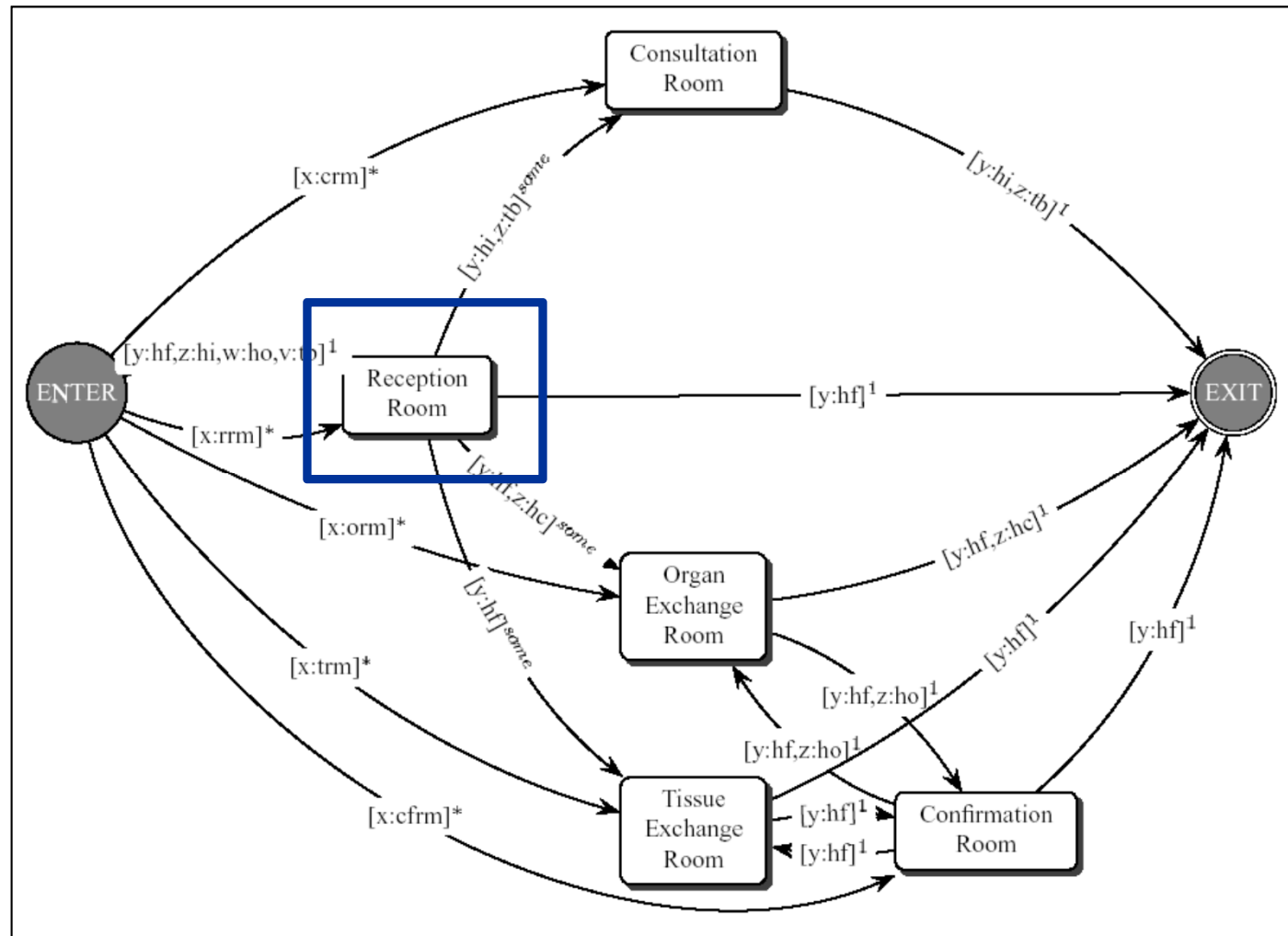
- AMELI is an institution middleware that is based in a formal electronic institution specification tool (ISLANDER), developed at IIIA
  - The ISLANDER framework is composed of:
    - A **Dialogical Framework**
      - Linguistic and social structure (**roles**) to give meaning to agent interactions, communication language
    - A **Performative Structure**
      - **scenes** and relationships between scenes (e.g. precedence)
    - **Rules**
      - Conventions to be followed, social commitments
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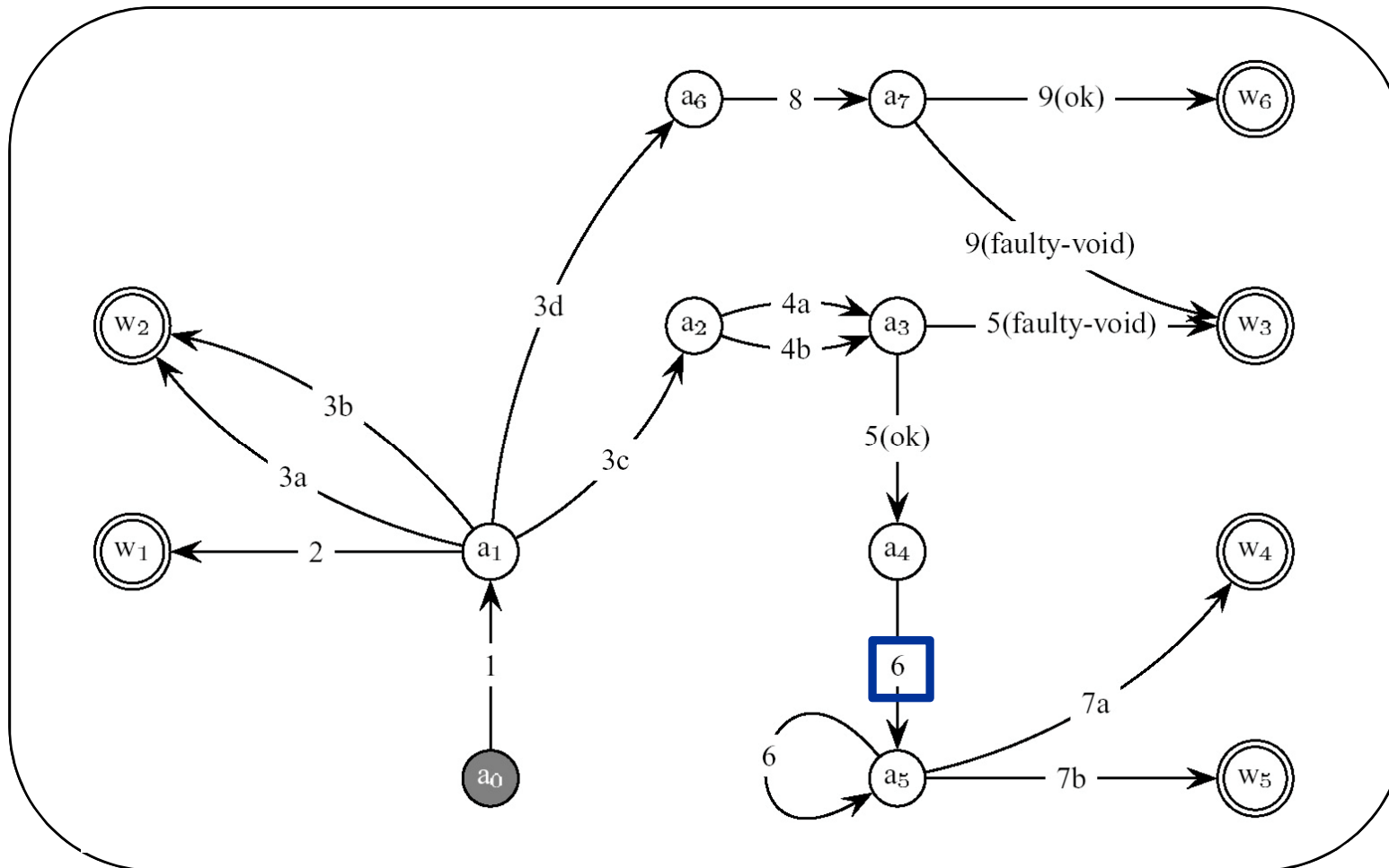
## AMELI (II)

- Two hypotheses:
    - All agent actions are **messages**, observable by the e-institution
    - An agent should **never** break the norms
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# ISLANDER: Performative Structure



# Scene conversational graph: Reception Room



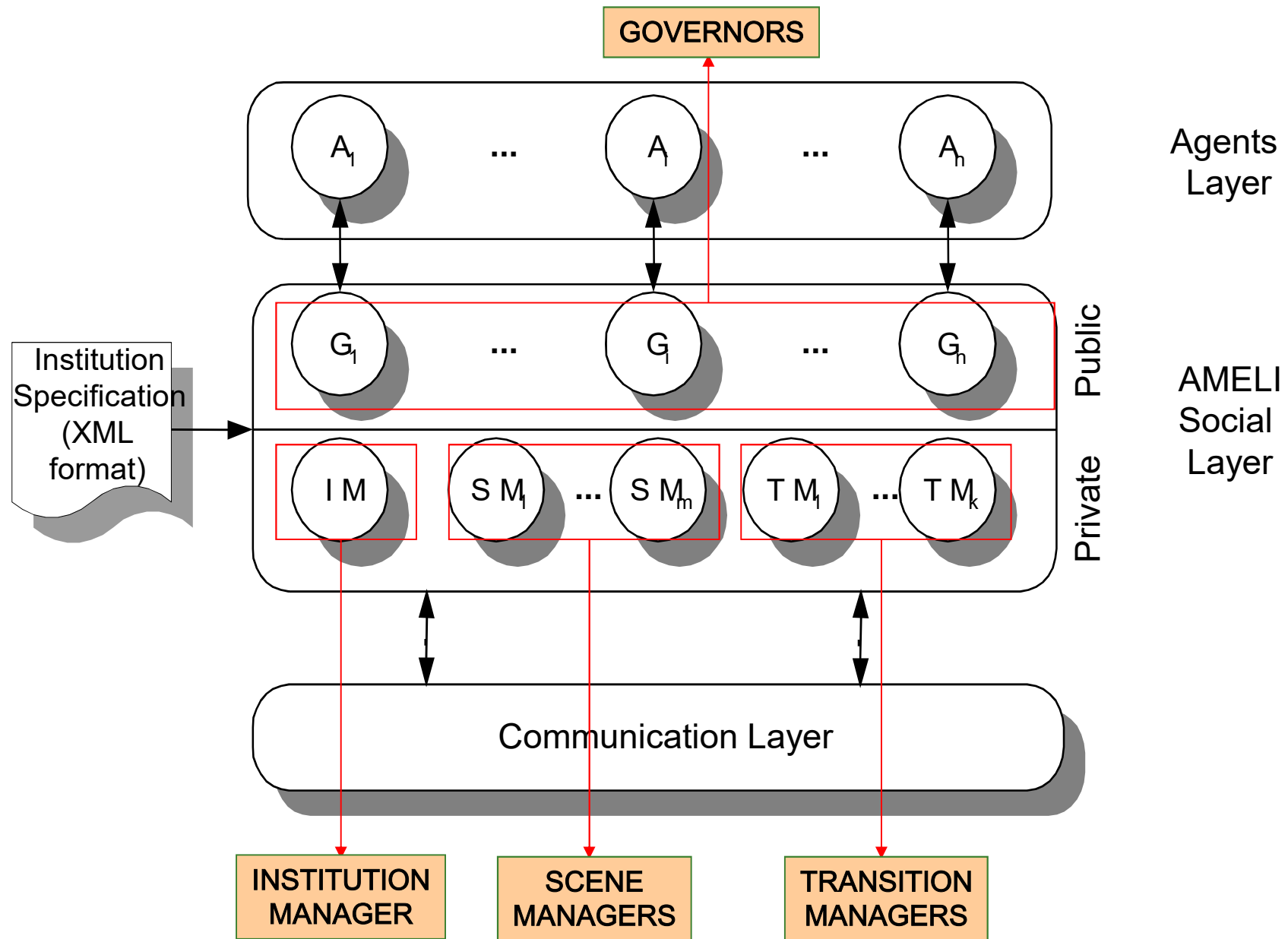
Each arrow is a concrete message

| Msg# | Illocution                                                                                                                                          |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | (request (?x hf hc tb) (?y rrm)<br>(admission ?id_agent ?role<br>?hospital_certificate))                                                            |
| 2    | (deny (!y rrm) (!x hf hc tb) (deny<br>?deny_reason))                                                                                                |
| 3a   | (accept (!y rrm) (!x hi) (accept_hi))                                                                                                               |
| 3b   | (accept (!y rrm) (!x tb) (accept_tb))                                                                                                               |
| 3c   | (accept (!y rrm) (!x hf) (accept_hf))                                                                                                               |
| 3d   | (accept (!y rrm) (!x hc) (accept_hc))                                                                                                               |
| 4a   | (inform (?x hf) (?y rrm) (petition_tissue<br>?id_hospital ?urgency_level<br>?time_to_deliver ?piece_type<br>(?piece_parameters) (?info_recipient))) |
| 4b   | (inform (?x hf) (?y rrm) (petition_organ<br>?id_hospital ?time_for_availability<br>?piece_type (?piece_parameters)<br>(?info_donor)))               |
| 5    | (inform (!y rrm) (!x hf) (petition_state<br>?id_petition ok faulty))                                                                                |
| 6    | (inform (?y rrm) (?x hf) (init_exchange<br>?piece_type ?id_exchange_room))                                                                          |
| 7a   | (request (?x hf) (?y rrm)<br>(tissue_exchange_entrance_request<br>!id_exchange_room))                                                               |
| 7b   | (request (?x hf) (?y rrm)<br>(organ_exchange_entrance_request<br>!id_exchange_room))                                                                |
| 8    | (inform (?x hc) (?y rrm)<br>(called_for_organ ?id_hospital<br>!id_petition))                                                                        |
| 9    | (inform (!y rrm) (!x hf) (called_state<br>!id_petition ok faulty))                                                                                  |

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## Objective of the AMELI middleware

- ❑ To mediate and facilitate agent communication within conversations (scenes)
  - ❑ To guarantee the correct evolution of each conversation (preventing errors made by the participating agents by filtering erroneous illocutions, thus protecting the institution)
  - ❑ To guarantee that agents' movements between scenes comply with the specification
  - ❑ To control which obligations participating agents acquire and fulfil
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## AMELI – Agents in Social Layer

- ❑ An *institution manager* that starts the institution, authorises agents to enter, and controls the creation of scenes
  - ❑ *Scene managers* responsible for governing scenes (one for scene)
  - ❑ *Transition managers* control agents' movements between scenes (one for transition)
  - ❑ *Governors* mediate the interaction of an agent with the rest of the agents within the institution and control the agents' obligations (one for participating agent)
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# Organisational Structures

- Define a pattern of information and control relationships between individuals, which **shapes the types of interactions** among them
  - Aids coordination by **specifying which actions an agent will undertake**
  - Social structure-based methods impose **restrictions** or **norms** on the behaviour of agents in a certain environment
    - avoid many potential conflicts, or ease their resolution
    - make it easier for a given agent to foresee and model other agents' behaviour in a closed environment and fit its own behaviour accordingly
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# Sociology and Societies

- The aim of any **society** is to allow its members to coexist in a shared environment and pursue their respective goals in co-operation with others
  - **Sociology** is a discipline that results from an evolution of Philosophy in order to describe the **interactions** that arise among the members of a group, and the **social structures** that are established
  - This can also be applied to digital societies composed by computational entities (**agent societies**)
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# Organisational studies

- Academic study of *organisations*, using the methods of *economics*, *sociology*, *political science*, *anthropology* and *psychology*
  - Concepts, abstractions and techniques coming from *organisational theories* and *organisational design* have been used in MAS
    - *Organisation theory* is a descriptive discipline, mainly focusing on *explaining* and understanding how organisations work
    - *Organisation design* is a normative, design-oriented discipline that aims to produce the frameworks and tools required to *create* effective organisations
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# Organisation design

- Organisation design involves the creation of roles, communication processes and formal reporting relationships in an organization
  - One can distinguish between two phases in an organisation design process:
    - *Strategic grouping*, which establishes the overall structure of the organisation (its main sub-units and their relationships), and
    - *Operational design*, which defines the more detailed roles and processes
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# Social Strucs. - Organisational classification

- Three main kinds of organisations:
    - Markets
    - Networks
    - Hierarchies
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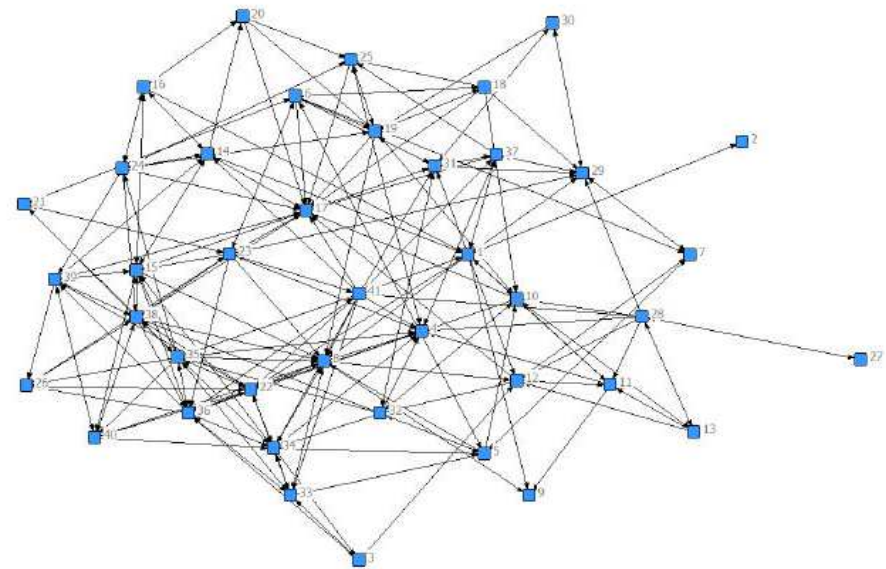
# Market structures

- They are well-suited for environments where the main purpose is the **exchange** of some goods
  - There are agents that provide services, agents that require services (and pay for them), and intermediate agents
  - All these self-interested agents are involved in **negotiations**
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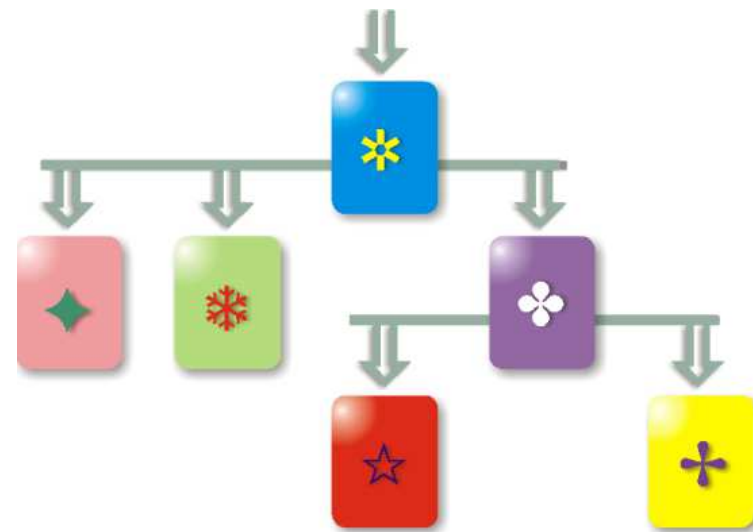
# Network structures

- They are well-suited for environments where (dynamic) *collaboration* among parties is needed
- There are *contracts* established in a dynamic way between the agents of the system
- *Coordination* is achieved by mutual interest



# Hierarchies

- Hierarchical structures are well-suited for environments where the society's purpose is the efficient *production* of some kind of results or goods
- Agents are specialised in concrete tasks and fully collaborative
- *Coordination* is achieved through *command and control* lines in a closed society



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# Kinds of social structures

|                            | <b>Market</b>                  | <b>Network</b>                   | <b>Hierarchy</b>          |
|----------------------------|--------------------------------|----------------------------------|---------------------------|
| <i>Type of society</i>     | Open                           | Trust                            | Closed                    |
| <i>Coordination</i>        | Price mechanism                | Collaboration                    | Supervision               |
| <i>Relation form</i>       | Competition                    | Mutual Interest                  | Authority                 |
| <i>Tone or climate</i>     | Precision/ suspi-<br>cion      | Open-ended/ mu-<br>tual benefits | Formal/ bureau-<br>cratic |
| <i>Conflict Resolution</i> | Haggling (Resort<br>to courts) | Reciprocity<br>(Reputation)      | Supervision               |

Table 1.1: V. Dignum *et al.* classification of Social Structures.

This classification is useful at the design stage, as it tries to motivate the choice of one structure based on its appropriateness for a specific environment

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# Social abstractions (I) - Role

- **Roles** identify activities and services necessary to achieve social objectives and enable to abstract from the specific individuals that will eventually perform them
    - From the *society design perspective*, roles provide the building blocks for the agent systems that can perform the role
    - From the *agent design perspective*, roles specify the expectations of the society with respect to the agent's activity in the society
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## Social abstractions (II) : Role Dependency

- Role *dependency* between two roles means that one role is dependent on another role for the realization of its objectives.
    - Societies establish *dependencies* and *power relations* between roles, indicating relationships between roles
    - These relationships describe how actors can interact and contribute to the realization of the objectives of each other. That is, an objective of a role can be delegated to, or requested from, other roles
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# Agent Societies – Models

- Formal **role models** reflect social competence of agents (rights and obligations) and allow to ensure a global behaviour of the system while also preserving individual flexibility
  - **Interaction models** reflect workflows and business processes
    - Explicit procedures and access requirements
    - Scene descriptions are formally specified, which allows verification
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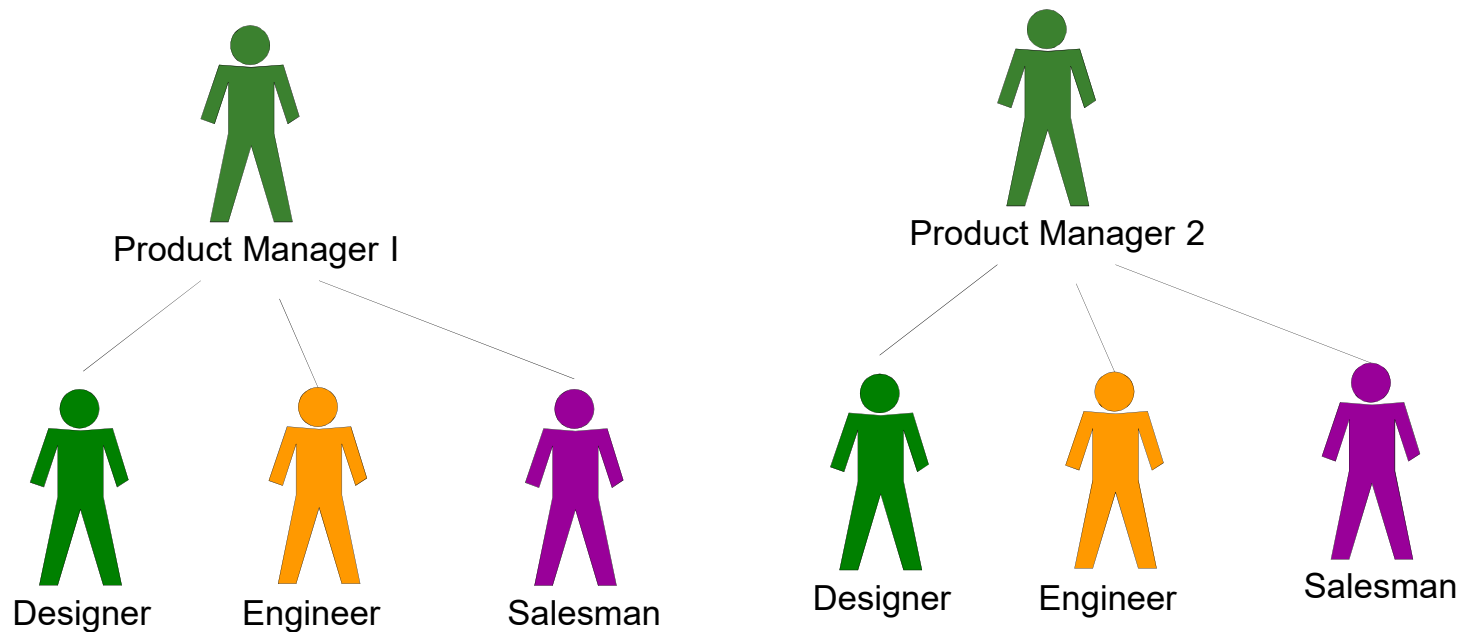
# Example of organisation structure

- Production of different types of cars within a factory
  - It involves several kinds of actors: engineers, designers, salesmen, different types of managers
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# Coordination Structure 1

## Product Hierarchy



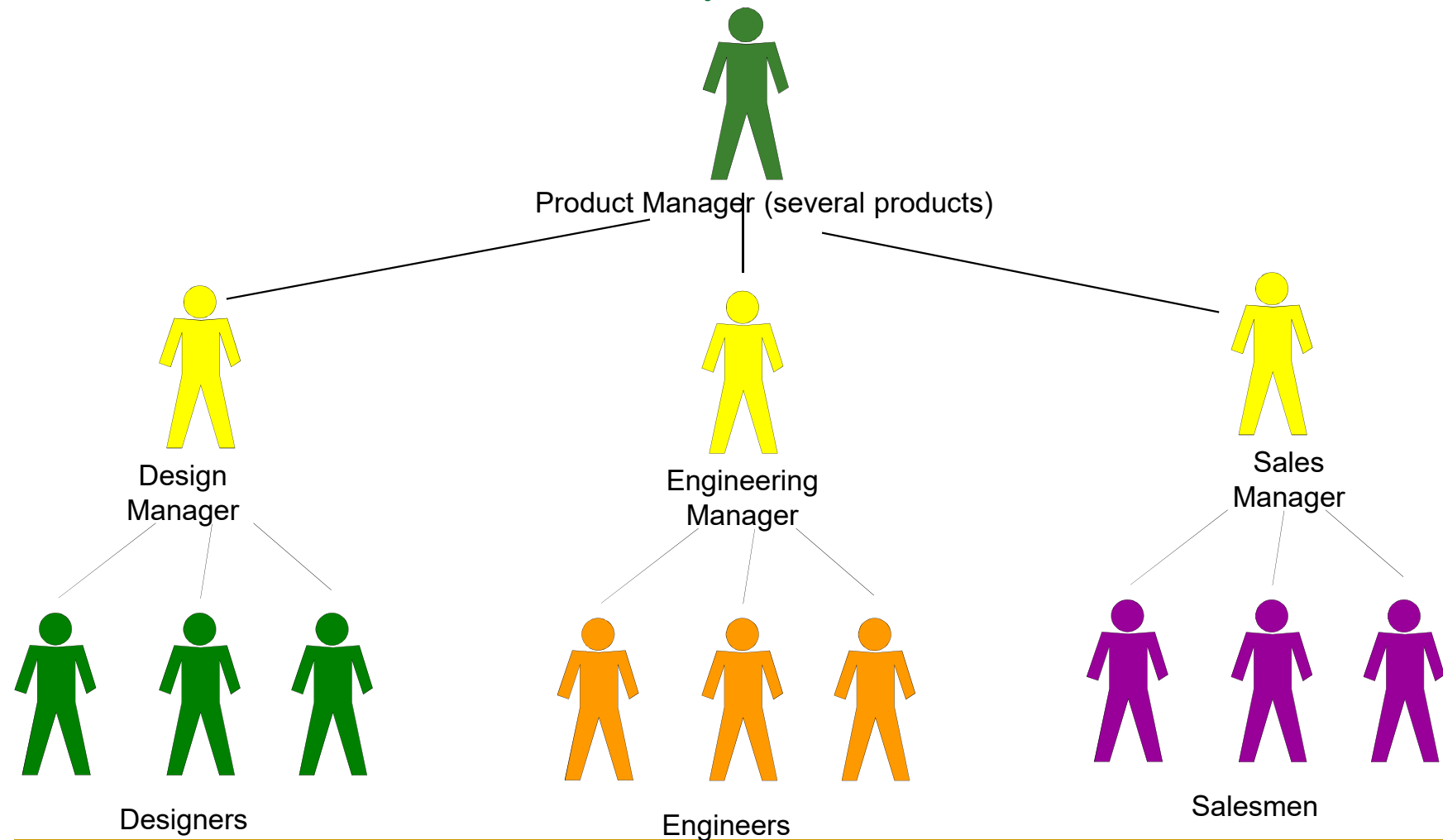
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# Product hierarchy

- There is a **dedicated team** for each product (type of car) to be produced
  - **Easy coordination** within each product team
  - There may be **global inefficiencies**
    - Repetition of design and engineering tasks
    - A salesman may be specialised in a single product, without enough knowledge to talk to a customer, identify his requirements and suggest the best product for him
    - There might be a “**global manager**” trying to provide some global communication and coordination
  - It might be a good option if products are quite different from each other
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# Coordination Structure 2

## Functional Hierarchy



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# Functional hierarchy (I)

- Actors with the same role work together under the supervision of a manager
- A **general product manager** coordinates all the activities of all the departments



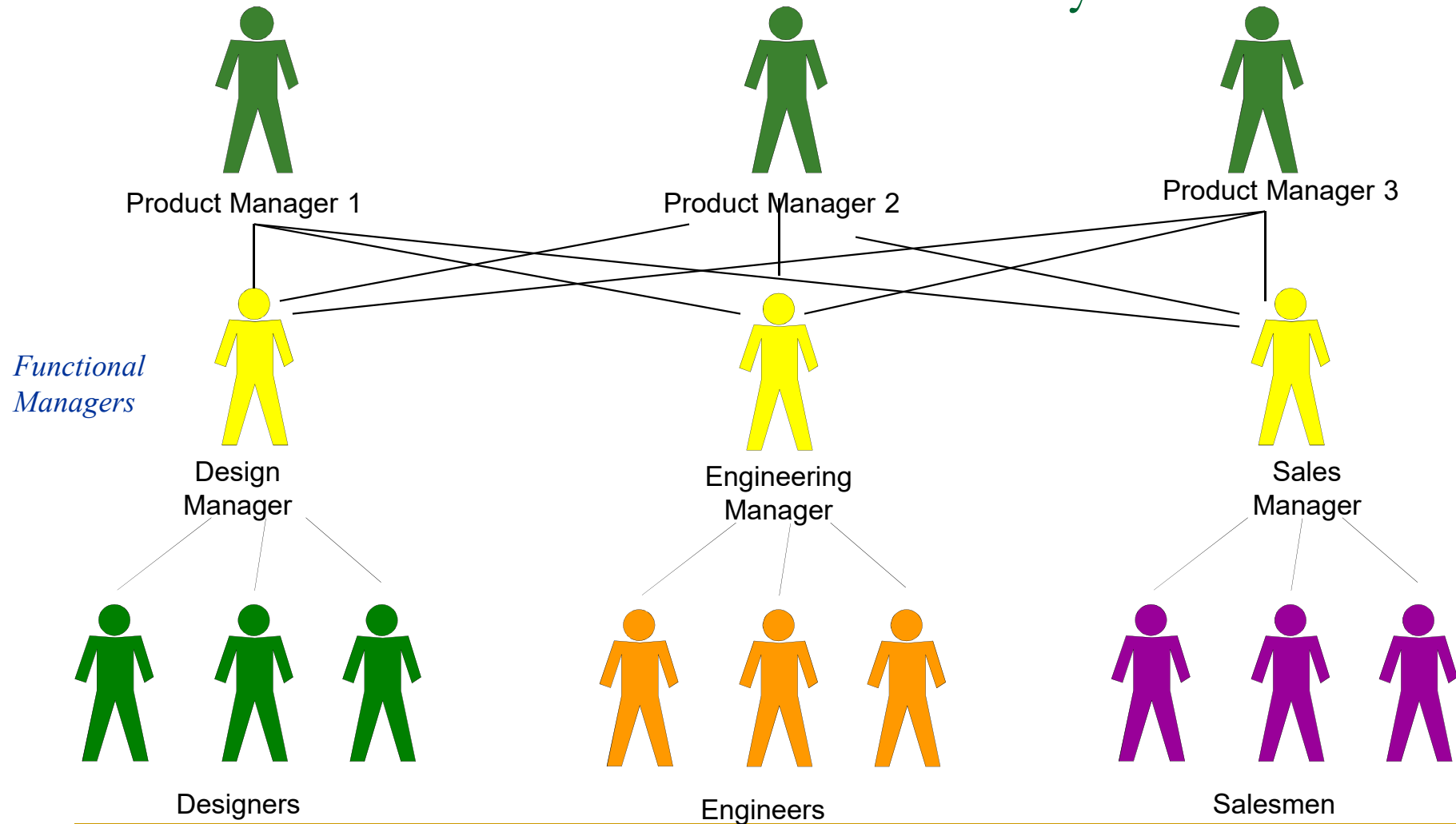
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## Functional hierarchy (II)

- The specialised actors can work in tasks **reusable in different products** (e.g. designing and engineering the air-conditioning system)
  - The resources in each department can be easily shared by its members
  - Much work concentrated in the global product manager, who must supervise the work of the whole system
  - It can be a good option if the different products are very interrelated
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# Coordination Structures 3

## Product + Functional Hierarchy



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# Product and functional hierarchy (I)

- There are specialised departments, with a manager for each of them (department head, or **functional manager**)
  - There is a **product manager for each product**, who talks to the functional managers
  - Functional managers act like brokers
  - Brokers are in contact with possible "workers" and will choose the best for each task
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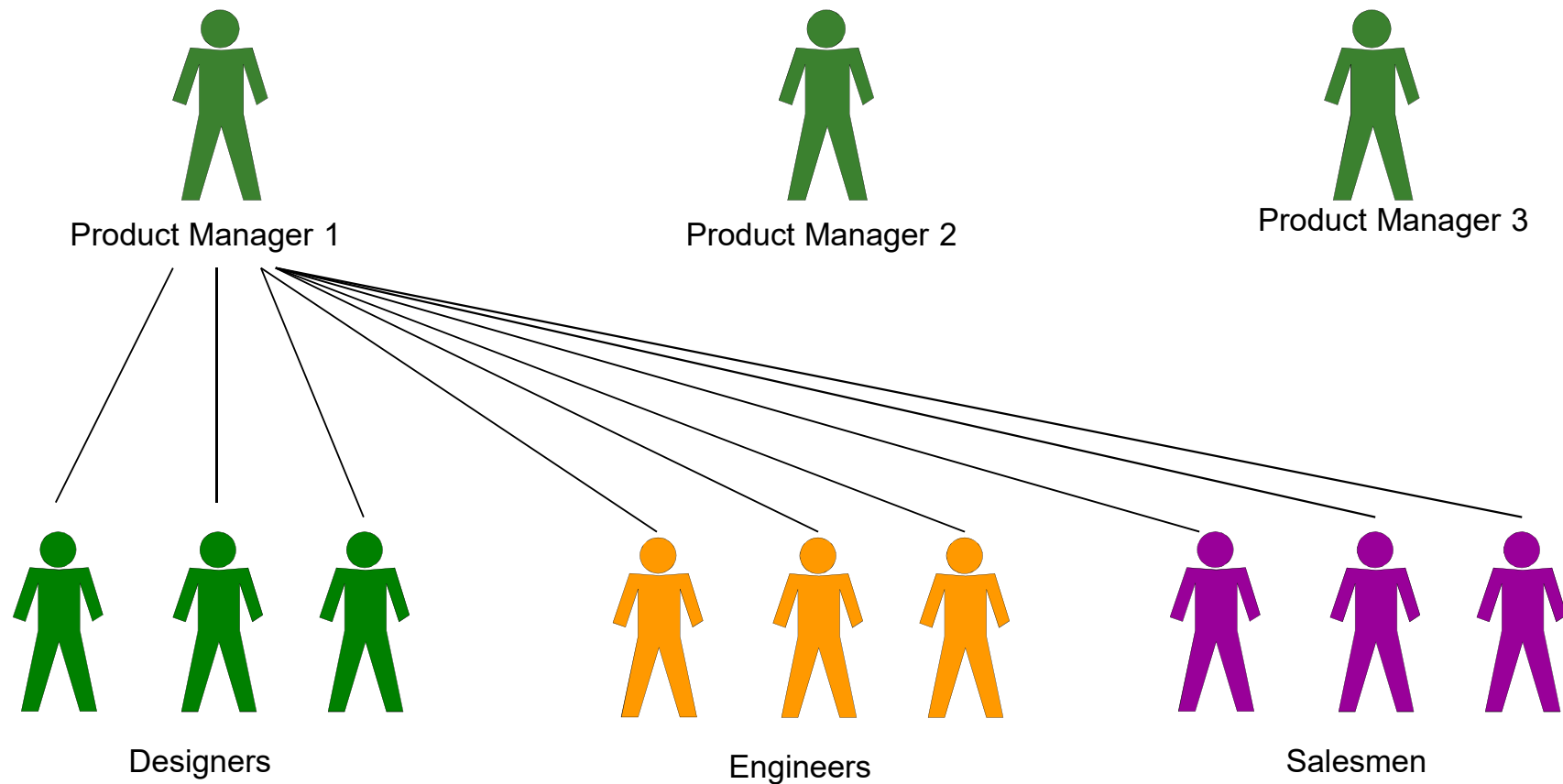
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## Product and functional hierarchy (II)

- Few connections and communication messages are required
  - Quite similar to the functional model
  - A lot of work for **functional managers**
    - Receive requests from several product managers
    - Coordinate the work of a team of agents
    - Identify common subtasks, manage shared resources
  - The failure of one product manager does not affect the others
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# Coordination Structure 4

## Flat Structure



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## Flat structure

- There is a product manager for each product, who talks directly to the low-level workers
  - A product manager may have to communicate with many different agents, and these agents have different abilities/expertise/vocabulary
  - Furthermore, there may be **inefficiencies** in the global behaviour
    - A designer could have work in 2 products, while another designer does not have any work
    - Two engineers could be working in similar problems in two different products
    - Difficult to solve even with a high-level global coordinator
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# Organisational Structures - Critique

- Useful when there are **master/slave** relationships in the MAS
    - However, the control over the slaves actions can mitigate some **benefits of DAI** such as reliability, concurrency
  - In some cases it presumes that at least **one agent has global overview** – an unrealistic assumption in MAS
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# Summary of Organisations

- *Focus on a structure / context for coordination*
  - Consider different types of structures:
    - Peer systems, markets, hierarchies, etc.
  - Are concerned with streamlining or “hard-wiring” certain patterns which help coordination in distributed problem solving
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## Readings for this week

- Sections 8.6.3/4 of the book *An introduction to MultiAgent Systems* (M. Wooldridge, 2nd edition)
  - Article: *Ant-based load balancing in telecommunications networks*
  - Article: *AMELI: An agent-based middleware for electronic institutions*
  - Article: *The organ allocation process: a natural extension of the Carrel agent-mediated electronic institution*
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